

Worksheet: 2-Input, 3-Hidden Node Neural Network

SECTION 1: Question Setup & Parameters

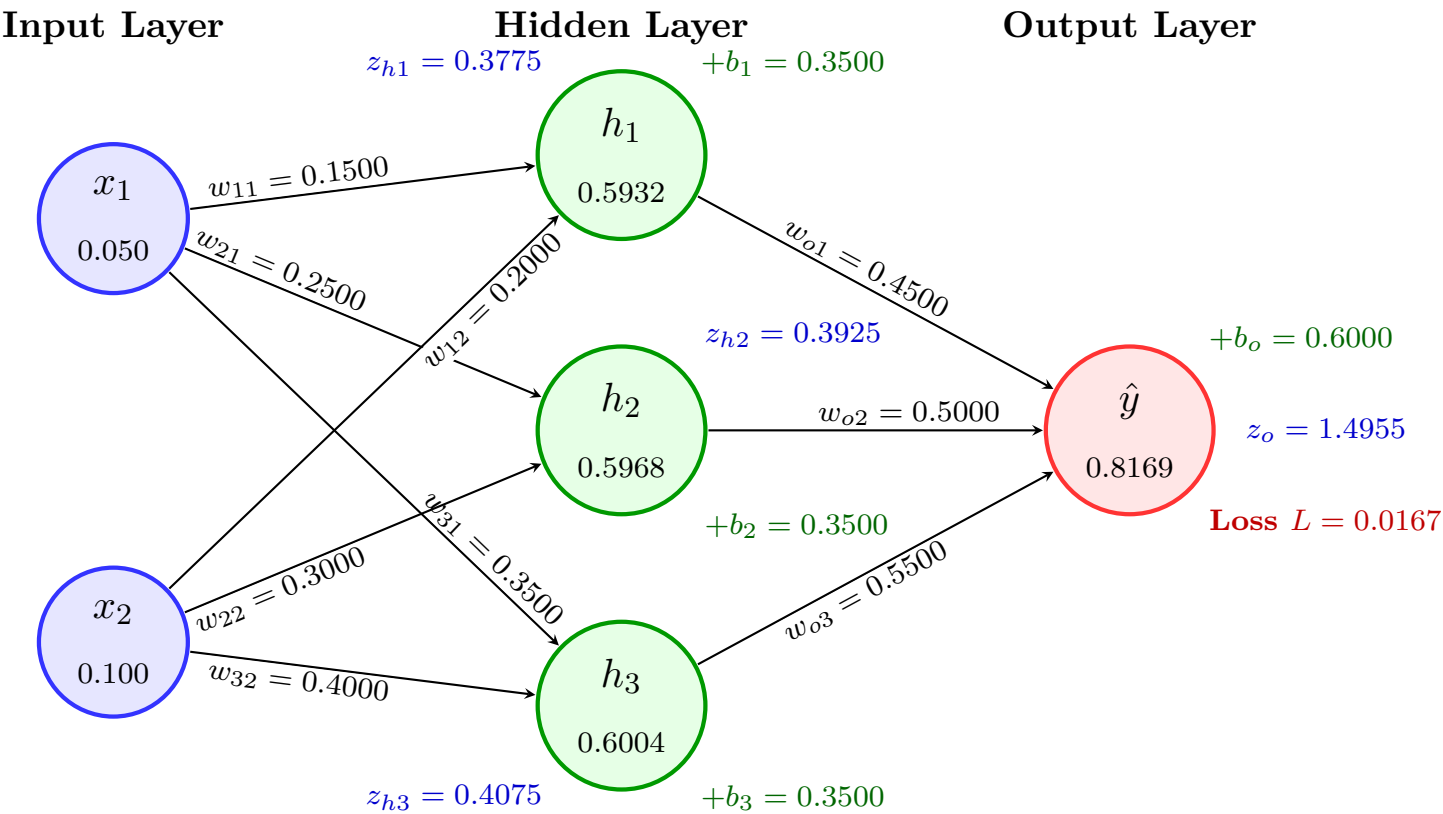
Problem Setup & Global Parameters:

- **Input Vector:** $\mathbf{x} = [x_1, x_2]^T = [0.050, 0.100]^T$
- **Target Output:** $y = 1.000$
- **Learning Rate:** $\eta = 0.500$
- **Activation Function:** Sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$
- **Activation Derivative:** $\sigma'(z) = \sigma(z)(1 - \sigma(z))$
- **Loss Function:** Squared Error $L = \frac{1}{2}(y - \hat{y})^2$
- **Loss Derivative:** $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$
- **Calculation Precision:** Fix all intermediate and final calculations at **4 decimal places** by truncation (cut off after 4th decimal digit without rounding).

Initial Network Weights & Biases:

- **Input $\rightarrow$ Hidden Weights ($W^{(1)}$):**
 $w_{11} = 0.1500, \quad w_{12} = 0.2000$
 $w_{21} = 0.2500, \quad w_{22} = 0.3000$
 $w_{31} = 0.3500, \quad w_{32} = 0.4000$
- **Hidden Biases:** $b_1 = 0.3500, \quad b_2 = 0.3500, \quad b_3 = 0.3500$
- **Hidden $\rightarrow$ Output Weights ($W^{(2)}$):**
 $w_{o1} = 0.4500, \quad w_{o2} = 0.5000, \quad w_{o3} = 0.5500$
- **Output Bias:** $b_o = 0.6000$

Neural Network Architecture Diagram (Initial Weights & First Forward Pass)



SECTION 2: First Forward Pass Calculations**2.1 Hidden Node Net Inputs (z_{hi}) and Activations (h_i):**

$$(a) \quad z_{h1} = w_{11}x_1 + w_{12}x_2 + b_1 = \underline{0.3775} \quad \Rightarrow \quad h_1 = \sigma(z_{h1}) = \underline{0.5932}$$

$$(b) \quad z_{h2} = w_{21}x_1 + w_{22}x_2 + b_2 = \underline{0.3925} \quad \Rightarrow \quad h_2 = \sigma(z_{h2}) = \underline{0.5968}$$

$$(c) \quad z_{h3} = w_{31}x_1 + w_{32}x_2 + b_3 = \underline{0.4075} \quad \Rightarrow \quad h_3 = \sigma(z_{h3}) = \underline{0.6004}$$

2.2 Output Unit Prediction ($\hat{y}$) and Initial Loss (L):

$$(a) \quad z_o = w_{o1}h_1 + w_{o2}h_2 + w_{o3}h_3 + b_o = \underline{1.4955} \quad \Rightarrow \quad \hat{y} = \sigma(z_o) = \underline{0.8169}$$

$$(b) \quad \text{Initial Loss } L = \frac{1}{2}(y - \hat{y})^2 = \underline{0.0167}$$

SECTION 3: Backward Pass & Update ALL Weights and Biases ($\eta = 0.500$)**3.1 Output Layer Error Gradient ($\delta_o = \frac{\partial L}{\partial z_o}$):**

$$\text{Formula: } \delta_o = \frac{\partial L}{\partial z_o} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_o} = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y})$$

$$(a) \quad \delta_o = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y}) = (0.8169 - 1.000) \cdot 0.8169(1 - 0.8169) = \underline{-0.0273}$$

3.2 Output Layer Updates (w_{oi}^+ and b_o^+):

$$\text{Formulas: } w_{oi}^+ = w_{oi} - \eta \frac{\partial L}{\partial w_{oi}} = w_{oi} - \eta(\delta_o \cdot h_i), \quad b_o^+ = b_o - \eta \frac{\partial L}{\partial b_o} = b_o - \eta \delta_o$$

$$(a) \quad w_{o1}^+ = w_{o1} - \eta(\delta_o \cdot h_1) = 0.4500 - 0.500(-0.0273 \cdot 0.5932) = \underline{0.4580}$$

$$(b) \quad w_{o2}^+ = w_{o2} - \eta(\delta_o \cdot h_2) = 0.5000 - 0.500(-0.0273 \cdot 0.5968) = \underline{0.5081}$$

$$(c) \quad w_{o3}^+ = w_{o3} - \eta(\delta_o \cdot h_3) = 0.5500 - 0.500(-0.0273 \cdot 0.6004) = \underline{0.5581}$$

$$(d) \quad b_o^+ = b_o - \eta(\delta_o) = 0.6000 - 0.500(-0.0273) = \underline{0.6136}$$

3.3 Hidden Layer Error Gradients ($\delta_{hi} = \frac{\partial L}{\partial z_{hi}}$):

$$\text{Formula: } \delta_{hi} = \frac{\partial L}{\partial z_{hi}} = (\delta_o \cdot w_{oi}) \cdot \sigma'(z_{hi}) = (\delta_o \cdot w_{oi}) \cdot h_i(1 - h_i)$$

$$(a) \quad \delta_{h1} = (\delta_o \cdot w_{o1}) \cdot h_1(1 - h_1) = (-0.0273 \cdot 0.4500) \cdot 0.5932(1 - 0.5932) = \underline{-0.0029}$$

$$(b) \quad \delta_{h2} = (\delta_o \cdot w_{o2}) \cdot h_2(1 - h_2) = (-0.0273 \cdot 0.5000) \cdot 0.5968(1 - 0.5968) = \underline{-0.0032}$$

$$(c) \quad \delta_{h3} = (\delta_o \cdot w_{o3}) \cdot h_3(1 - h_3) = (-0.0273 \cdot 0.5500) \cdot 0.6004(1 - 0.6004) = \underline{-0.0035}$$

3.4 Input-to-Hidden Layer Updates (w_{ij}^+ and b_i^+):

$$\text{Formulas: } w_{ij}^+ = w_{ij} - \eta \frac{\partial L}{\partial w_{ij}} = w_{ij} - \eta(\delta_{hi} \cdot x_j), \quad b_i^+ = b_i - \eta \frac{\partial L}{\partial b_i} = b_i - \eta \delta_{hi}$$

$$(a) \quad w_{11}^+ = w_{11} - \eta(\delta_{h1} \cdot x_1) = 0.1500 - 0.500(-0.0029 \cdot 0.050) = \underline{0.1500}$$

$$(b) \quad w_{12}^+ = w_{12} - \eta(\delta_{h1} \cdot x_2) = 0.2000 - 0.500(-0.0029 \cdot 0.100) = \underline{0.2001}$$

$$(c) \quad w_{21}^+ = w_{21} - \eta(\delta_{h2} \cdot x_1) = 0.2500 - 0.500(-0.0032 \cdot 0.050) = \underline{0.2500}$$

$$(d) \quad w_{22}^+ = w_{22} - \eta(\delta_{h2} \cdot x_2) = 0.3000 - 0.500(-0.0032 \cdot 0.100) = \underline{0.3001}$$

$$(e) \quad w_{31}^+ = w_{31} - \eta(\delta_{h3} \cdot x_1) = 0.3500 - 0.500(-0.0035 \cdot 0.050) = \underline{0.3500}$$

$$(f) \quad w_{32}^+ = w_{32} - \eta(\delta_{h3} \cdot x_2) = 0.4000 - 0.500(-0.0035 \cdot 0.100) = \underline{0.4001}$$

$$(g) \quad b_1^+ = b_1 - \eta(\delta_{h1}) = 0.3500 - 0.500(-0.0029) = \underline{0.3514}$$

$$(h) \quad b_2^+ = b_2 - \eta(\delta_{h2}) = 0.3500 - 0.500(-0.0032) = \underline{0.3516}$$

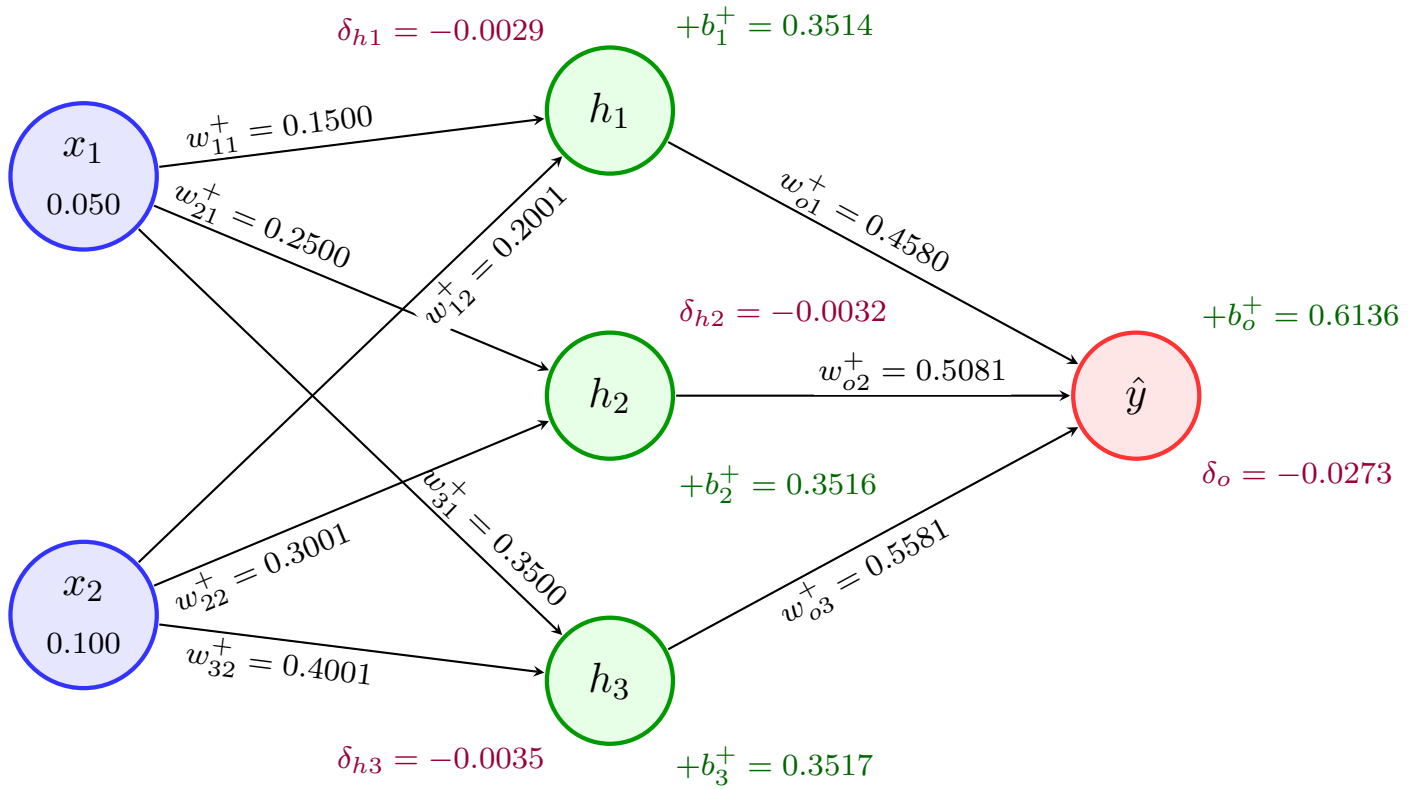
$$(i) \quad b_3^+ = b_3 - \eta(\delta_{h3}) = 0.3500 - 0.500(-0.0035) = \underline{0.3517}$$

Updated Neural Network Architecture (Gradients & New Weights/Biases)

Input Layer

Hidden Layer

Output Layer



SECTION 4: Second Forward Pass (Verification of Learning)

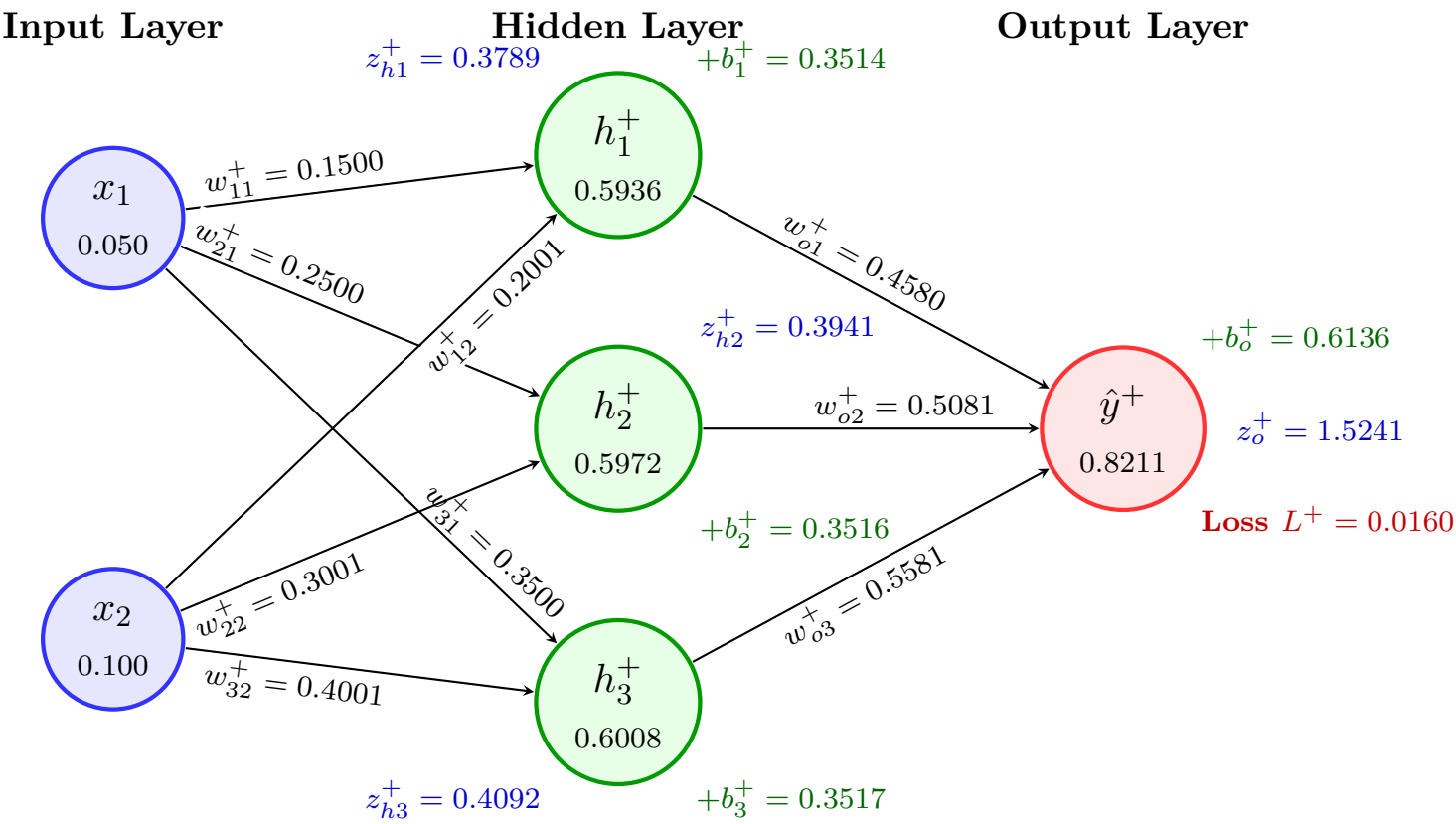
4.1 Recalculate Hidden Activations (h_i^+):

- (a) $z_{h1}^+ = w_{11}^+x_1 + w_{12}^+x_2 + b_1^+ = \underline{0.3789} \Rightarrow h_1^+ = \sigma(z_{h1}^+) = \underline{0.5936}$
- (b) $z_{h2}^+ = w_{21}^+x_1 + w_{22}^+x_2 + b_2^+ = \underline{0.3941} \Rightarrow h_2^+ = \sigma(z_{h2}^+) = \underline{0.5972}$
- (c) $z_{h3}^+ = w_{31}^+x_1 + w_{32}^+x_2 + b_3^+ = \underline{0.4092} \Rightarrow h_3^+ = \sigma(z_{h3}^+) = \underline{0.6008}$

4.2 Recalculate Output Prediction ($\hat{y}^+$) and New Loss (L^+):

- (a) $z_o^+ = w_{o1}^+h_1^+ + w_{o2}^+h_2^+ + w_{o3}^+h_3^+ + b_o^+ = \underline{1.5241} \Rightarrow \hat{y}^+ = \sigma(z_o^+) = \underline{0.8211}$
- (b) New Loss $L^+ = \frac{1}{2}(y - \hat{y}^+)^2 = \underline{0.0160}$
- (c) Has the loss decreased after gradient descent update? ($L^+ < L$): **Yes ($\underline{0.0160} < \underline{0.0167}$)**

Neural Network Architecture Diagram (Second Forward Pass with Updated Weights)



SECTION 5: Parameter Updates in Backpropagation

5.1 All Parameters Must Be Updated

- The learnable parameters include both weights and biases:

$$\theta = \{\mathbf{W}^{(1)}, \mathbf{b}^{(1)}, \mathbf{W}^{(2)}, \mathbf{b}^{(2)}\}.$$

- Gradient descent updates every parameter:

$$\theta^+ = \theta - \eta \nabla_{\theta} L.$$

- Weights** control the strength of connections, while **biases** shift the activation/decision boundary.

5.2 Backpropagation Order & Role of the Loss Derivative

Step 1: Output Error Gradient (δ_o) and the Loss Derivative: By the multivariate chain rule, the error gradient at the output net input z_o decomposes into:

$$\delta_o = \frac{\partial L}{\partial z_o} = \underbrace{\frac{\partial L}{\partial \hat{y}}}_{\text{Loss Derivative}} \cdot \underbrace{\frac{\partial \hat{y}}{\partial z_o}}_{\text{Activation Derivative}} = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y}).$$

- Loss Derivative** $\frac{\partial L}{\partial \hat{y}}$: Differentiating squared error $L = \frac{1}{2}(y - \hat{y})^2$ with respect to $\hat{y}$ gives $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$.
In our problem: $\frac{\partial L}{\partial \hat{y}} = 0.8169 - 1.000 = -\mathbf{0.1831}$.
- Activation Derivative** $\frac{\partial \hat{y}}{\partial z_o}$: Differentiating sigmoid $\hat{y} = \sigma(z_o)$ with respect to z_o gives $\sigma'(z_o) = \hat{y}(1 - \hat{y}) = 0.8169(1 - 0.8169) = \mathbf{0.1495}$.
- Why it is combined in Question 3.1:** The question combines them into $\delta_o = (-0.1831)(0.1495) = -\mathbf{0.0273}$. The loss derivative is thus the initial driving signal for all downstream parameter updates.

Step 2: Update output parameters:

$$w_{oi}^+ = w_{oi} - \eta(\delta_o h_i), \quad b_o^+ = b_o - \eta \delta_o.$$

Step 3: Hidden errors (propagated backwards via chain rule):

$$\delta_{hi} = \frac{\partial L}{\partial z_{hi}} = (\delta_o w_{oi}) h_i(1 - h_i).$$

Step 4: Update hidden parameters:

$$w_{ij}^+ = w_{ij} - \eta(\delta_{hi} x_j), \quad b_i^+ = b_i - \eta \delta_{hi}.$$

5.3 Important Rules

- Role of the Loss Derivative:** Changing the loss function (e.g., from Squared Error to Cross-Entropy) only alters the initial loss derivative $\frac{\partial L}{\partial \hat{y}}$ inside δ_o ; the rest of the backpropagation chain rule remains identical.
- Update both weights and biases** across all layers.
- Bias gradient:** $\frac{\partial L}{\partial b} = \delta$ because $\frac{\partial z}{\partial b} = 1$.
- Weight gradient:** $\frac{\partial L}{\partial w} = \delta \cdot \text{input}$.
- Use pre-update weights** when propagating δ to previous layers.
- Backpropagation proceeds from **output** $\rightarrow$ **hidden** $\rightarrow$ **input**.